Year 12 Maths - Coordinate geometry

Similar questions - Circles, triangles and touching circles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The points $A(-1,0)$, $B(3,2)$ and $C(4,0)$ are joined to form a triangle.

(a) Show that angle $ABC = 90^\circ$. **[3]**

(b) Hence find the equation of the circle that passes through A, B and C, explaining your method. **[4]**

(c) $ABCD$ is a rectangle. Find the coordinates of D and show that D lies on the circle. **[3]**

(d) Find the area of the rectangle $ABCD$. **[2]**

2 Circle C has equation $x^2 + y^2 - 2x - 4y - 4 = 0$. Circle D has radius 2, its centre lies on the line $y = x$, and it touches circle C externally (at exactly one point, with neither circle inside the other).

(a) Find the centre and radius of C. **[3]**

(b) Show that the x-coordinate t of the centre of D satisfies $t^2 - 3t - 10 = 0$. **[3]**

(c) Hence find the two possible centres of D. **[2]**

(d) For the circle D whose centre has positive coordinates, find the coordinates of the point where it touches C. **[3]**

Question 1

(a) $\overrightarrow{AB} = \binom{4}{2}$, $\overrightarrow{BC} = \binom{1}{-2}$. Gradients $\frac{1}{2}$ and -2 have product -1 (or dot product $4-4=0$), so $AB \perp BC$

(b) The angle in a semicircle is $90°$, so AC is a diameter. Centre $\left(\frac{3}{2}, 0\right)$, radius $\frac{5}{2}$: $\left(x-\frac{3}{2}\right)^2 + y^2 = \frac{25}{4}$

(c) $D = A + C - B = (0,-2)$. $\left(0-\frac{3}{2}\right)^2 + (-2)^2 = \frac{9}{4} + 4 = \frac{25}{4}$, so D is on the circle

(d) $AB = \sqrt{20}$, $BC = \sqrt{5}$, area $= \sqrt{100} = 10$

Question 2

(a) $(x-1)^2 + (y-2)^2 = 9$. Centre $(1,2)$, radius 3

(b) Touching externally: distance between centres $= 3+2 = 5$. $(t-1)^2 + (t-2)^2 = 25$, so $2t^2 - 6t + 5 = 25$, giving $t^2 - 3t - 10 = 0$

(c) $(t-5)(t+2) = 0$: centres $(5,5)$ and $(-2,-2)$

(d) The point lies on the line from $(1,2)$ to $(5,5)$, a fraction $\frac{3}{5}$ of the way: $(1,2) + \frac{3}{5}(4,3) = (3.4,\, 3.8)$